For the paragraph indentation feature, `parize` supports processing all block-level elements; for the paragraph spacing feature, we exclude the elements `block`, `pad`, `grid`, `stack`, and `layout`. This is because most Typst package (including Typst's own elements such as `list`, `terms`, etc.) use these elements to create containers and override their behavior; if `parize` were to support them, it could sometimes lead to layout confusion or be puzzling for users.

If you want a custom block-level container to have paragraph spacing and paragraph indentation properties (i.e., to be treated as part of a paragraph),

- you can mark it with `parize-par-above-flag` and `parize-par-below-flag` before and after the container.
- Or you can wrap it in `parize-block`.

When users apply `parize` to `block`, such containers will be processed as part of a paragraph.

- Alternatively, you can mark block-level elements with `parize-prevention-label` so that `parize` does not process them.

Definition of `parize-par-above-flag`, `parize-par-below-flag`

typst

```typst
#let parize-block-label = label("__cdl_parize_block_flag__")
#let parize-par-below-flag = [#block(below: auto, above: 0pt, width: 0pt, height: 0pt, none)#parize-block-label]
#let parize-par-above-flag = [#block(below: 0pt, above: auto, width: 0pt, height: 0pt, none)#parize-block-label]
```

Definition of `parize-block`

typst

```typst
#let parize-block(body, ..block-args) = {
  [#block(..block-args, body)#parize-block-label]
}
```

Definition of `parize-prevention-label`

typst

```typst
#let parize-prevention-label = label("__cdl_parize_prevent-label__")
```

Show-Rule Order

First, note the order of application issue with `par-indent`. Suppose an element is overridden by `show elem: ...`. If you want `par-indent` to apply to the overridden element, generally you should place `show elem: ...` after `show: par-indent`; for example:

Lorem ipsum dolor sit amet, consectetur. Test figure Lorem ipsum dolor sit amet, consectetur. Test: Figure 1. Lorem ipsum dolor sit amet, consectetur. Test figure Lorem ipsum dolor sit amet, consectetur.	Lorem ipsum dolor sit amet, consectetur. Test figure Lorem ipsum dolor sit amet, consectetur. Test: Figure 3. Lorem ipsum dolor sit amet, consectetur. Test figure Lorem ipsum dolor sit amet, consectetur.

Code:

```typst
#set par(first-line-indent: (amount: 2em, all: false), spacing: 1.5em)
#show "Test figure": set text(fill: red) // debug

#import "../lib.typ": *

#let test-figure(label-test: "fig:parize") = [
  #lorem(8)
  #figure()[
    #parize-par-above-flag
    Test figure
    #parize-par-below-flag
  ]#label(label-test)
  #lorem(10)

  Test: #ref(label(label-test)).

  #lorem(8)

  #figure()[
    #parize-par-above-flag
    Test figure
    #parize-par-below-flag
  ]

  #lorem(10)
]

#table(
  columns: (1fr,) * 2,
  [
    #show: par-indent.with(
      include-elem: (block,),
      use-par-leading: (
        block-text-leading: (block,),
        text-block-leading: (block, figure),
      ),
    )

    #show figure: it => it.body

    // `par-indent` receives `figure.body`, so it applies to `block`'s rules

    #test-figure(label-test: "fig:parize")

  ],
  [
    #show figure: it => it.body

    #show: par-indent.with(
      include-elem: (block,),
      use-par-leading: (
        block-text-leading: (block,),
        text-block-leading: (block, figure),
      ),
    )

    // `par-indent` recognizes `figure` as a block-level element, so it applies to `figure`'s rules

    #test-figure(label-test: "fig:parize2")
  ],
)
```

Theorem Environments

To make theorem environments provided by `theorion` be treated as part of a paragraph, you can do as follows:

typst
```
#import "@preview/theorion:0.6.0": *
#import cosmos.fancy: fancy-box // style

#let render = (prefix: none, title: "", full-title: auto, body) => {
  parize-par-above-flag // add
  fancy-box(
    prefix: prefix,
    title: title,
    full-title: full-title,
    body,
  )
  parize-par-below-flag // add
}
// theorem environment
#let (theorem-counter, theorem-box, theorem, show-theorem) = make-frame(
  "theorem",
  theorion-i18n-map.at("theorem"),
  inherited-levels: 2,
  render: render,
)

// let us define a proof environment, which is different from theorion's proof environment
#let proof(..args, body, title: none) = {
  let _title = args.pos().at(0, default: none)
  if title == none {
    title = if type(_title) == content { _title } else { [Proof] }
  }
  block(
    stroke: (left: orange + 2pt),
    inset: (left: 5pt),
    outset: (y: 2pt),
    width: 100%,
    {
      parize-par-above-flag // make sure the first line is not indented
      emph(title)
      [.]
      h(.5em, weak: true)
      body
      math.qed
    },
  )
  parize-par-below-flag
}
```

The function `render` can also be definded as follows:

typst
```
#let render2 = (prefix: none, title: "", full-title: auto, body) => {
  parize-block(
    fancy-box(
      prefix: prefix,
      title: title,
      full-title: full-title,
      body,
    ),
  )
}
```

Effect:

The Fundamental Theorem of Calculus is an extremely powerful theorem that establishes the relationship between differentiation and integration, and gives us a way to evaluate definite integrals without using Riemann sums or calculating areas.

Theorem 1 (The Fundamental Theorem of Calculus)

If $f(x)$ is continuous over an interval $[a, b]$ and $F(x)$ is any antiderivative of $f(x)$, then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

In particular, $\frac{\mathrm{d}}{\mathrm{d}x}\int_a^x f(t)\,\mathrm{d}t = f(x)$, $x \in [a, b]$.

Before we delve into the proof of Theorem 1, a couple of subtleties are worth mentioning here.
- Lorem ipsum.
- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

Lorem ipsum dolor sit amet, consectetur.

Now we can start the proof.

Proof of Theorem 1. Step I. Lorem ipsum dolor.

Step II. Lorem ipsum dolor. ∎

We say ... Lorem ipsum.

Proof of Theorem 1 (Second method). Step I. Lorem ipsum dolor. Applying

$$F'(x) = \lim_{h \to 0} \frac{F(x+h) - F(x)}{h}$$

Step II. Lorem ipsum dolor. ∎

Note that, ...

Code:

typst
```
#show link: set text(fill: orange)
#set par(first-line-indent: (amount: 2em, all: true), spacing: 1.5em)

#import "../lib.typ": *
#show: par-indent.with(
  include-elem: (block, math.equation, enum, list, terms),
  use-par-leading: (
    // apply-elem: "all",
    block-text-leading: (block, enum, list, terms),
    text-block-leading: (block, enum, list, terms),
  ),
)

#show: show-theorem // after `parize`: make sure all the elements in theorem are the overridden elements (Note that, in `theorion`, it overrides `figure` in `show-theorem`)

The Fundamental Theorem of Calculus is an extremely powerful theorem that establishes the relationship between differentiation and integration, and gives us a way to evaluate definite integrals without using Riemann sums or calculating areas.

#theorem[The Fundamental Theorem of Calculus][
  If $f(x)$ is continuous over an interval $[a, b]$ and $F(x)$ is any antiderivative of $f(x)$, then
  $
    integral_a^b f(x) dif x = F(b) - F(a).
  $
  In particular, $dif / (dif x) integral_a^x f(t) dif t = f(x)$, $x in [a, b]$.
]<thm:Newton-Leibniz>
Before we delve into the proof of @thm:Newton-Leibniz, a couple of subtleties are worth mentioning here.
- #lorem(2)
- #lorem(10)
#lorem(6)

Now we can start the proof.
#proof[Proof of @thm:Newton-Leibniz][
  Step I. #lorem(3)

  Step II. #lorem(3)
]

We say ... #lorem(2)

#proof[Proof of @thm:Newton-Leibniz (Second method)][
  Step I. #lorem(3) Applying
  $
    F'(x) = lim_(h->0) (F(x+h) - F(x)) / h
  $

  Step II. #lorem(3)
]
Note that, ...
```